

Research Proposal - Complexity Costs of Fault-Tolerance Circuits.

Michael Ben-Or David Ponnarovsky

January 6, 2026

Abstract

We aim to study the overhead costs of fault-tolerant circuits in terms of both depth and width expansions. This means we ask what the optimal cost factor is, in each of the parameters alone, that one can hope to achieve, and also raise the question about the trade-off relation between the two. We ask these questions from both classical and quantum perspectives. The following proposal formulates the problems and states subproblems that might shed light on the answer.

We focus on shallow depth circuit classes, in particular, $\mathbf{QAC}_0$, $\mathbf{QNC}_{0,f}$, $\mathbf{QNC}_1$, and 'QNC without reset gates', and certain known problem candidates for demonstrating quantum advantage such as factoring [Sho97] and Instantaneous Quantum Polynomial-time [BMS17], [Pal+24]. We aim to answer whether there exists a fault tolerance with only constant overhead at the circuit's depth. A positive answer implies that computing logarithmic depth circuits when subjected to noise is not harder than computing in an ideal environment.

1 Introduction.

The Quantum Computation model is widely believed to be superior to classical models, offering asymptotic speedup in tasks such as factoring [Sho97], searching [Gro96], simulating, and more relative to the best-known classical solutions. Yet even though there is almost complete agreement about the superiority of the ideal quantum model, there is still a debate over whether it is possible to implement complex computation in the real world, where the qubits and gates are subject to faults. Similarly, the feasibility of realizing classical computation has also been an open question. In fact the question about the feasibility of computation under noise is almost as ancient as the computer science field itself, initialized by Von Neumann [Neu56] at the time that classical computation putted in debuts. Time been pass and the followed works had pointed that not even a polynomial computation in the presence of noise is still reasonable but one can implement a fault tolerance version at a most constant times cost at the circuits depth [Pip85]. Or in asymptotic sense, classical computation in the presence of noise is as exactly hard as computation in ideal environment.

Recently, the feasibility question has been raised again, this time regarding quantum circuits, and while an intensive work has been done, and also succeed to prove that polynomial quantum computation can be made fault tolerance, [AB99], [Got14] and even with only constant overhead at the original circuit width [Gro19], polylogarithmic depth [TKY24], and logarithmic depth [NP24]. Yet the minimal required depth overhead is still not well understood. We stress that in all the mentioned constructions, the original constant-depth gates are mapped to logical presentations whose underlying implementation has asymptotically growing depth. This is in contrast to Pippenger [Pip85], where the logical gates are $O(1)$ -depth circuits.

Such an expansion prevents any construction that merely shifts presentations without interfering with the logical computation path of the original circuits from reaching a constant depth overhead. Previous works have made way to bypass this obstacle; notably, [Ngu24] and [Li+25] present good codes with transversal $\mathbf{CCZ}$ gates, and the series of works by Bombin [BM06], [Bom10], [Bom15] yield the 2D-color and 3D-color codes, which can be embedded on low-dimensional lattices (2D and 3D) and offer transversal gate implementation, where the first

offers the whole Clifford group and the second the **T** gate. However, no code with a non-trivial distance and $O(1)$ -depth implementation for a complete universal gate set is known.

Moreover, even the depth overhead is particularly interesting as today’s quantum machines are challenged to maintain quantum states for a long time. The limitations of these machines have motivated research to define NISQ, which stands for Noisy Intermediate-Scale Quantum, referring to the current era of quantum computing characterized by quantum processors that have a limited number of qubits and are prone to errors due to noise. In addition to NISQ, another common characterization for limited quantum computation is computation without reset gates, which has been proved to be impossible when restricted to polynomial space [Aha+96]. Having a constant depth-overhead fault tolerance scheme would imply the feasibility of log depth computation in that model.

This work addresses the above. We ask whether a magnitude depth overhead is an unavoidable price that one has to pay, If a Constant Depth Fault Tolerance Construction (CDFT) exists, and if so, how to construct it. In particular, whether an ideal $\mathbf{QNC}_1$, the class of problems that can be decided by logarithmic depth quantum circuits, can be computed in noisy- $\mathbf{QNC}_1$ circuits.

Additionally, we extend the question beyond the standard classes and ask about fault-tolerant sampling. Specifically, we consider several sampling processes by shallow quantum circuits, which are believed to be infeasible for classical circuits, such as Instantaneous Quantum Polynomial-time [BMS17], [Pal+24], and ask about the depth efficiency of their fault-tolerant version.

Classical Lower bounds [Gal91], [RS91]. [Rom06] shows that a tradeoff in model in which the given input is already encoded. Then the width overhead of the circuit is $\Theta(\frac{\log n}{\log \log n})$ while the depth overhead is $\Theta(\log \log n)$. [COMMENT] Re-write this. [KZ25], A work that studied width-depth tradeoff, and shows that codes with non-trivial rate and distance, can not offer the control-X gate for any pair of logical bits.

Something about the importance of fault tolerance computation to PCP construction. New classical [BMV24], and quantum construction [ABN24]. [PS94] Seems like a construction that based on the fact that the previous circuit was fault tolerance.

The proposal is organized as follows. ?? presents the notations, formal definitions, and states the open problems that will be studied through the research. Then, ?? describes strategies to prove CDFT. In particular, it lists primitives that can be used to achieve it and discusses how far we are from obtaining them. Having said that, ?? presents the first cues against the possibility of CDFT and provides the entry points to prove the impossibility claim. Finally, ?? discusses the applications and implications of either the correctness of CDFT or the impossibility of CDFT, from both theoretical and practical views.

References

- [Neu56] J. von Neumann. “Probabilistic Logics and the Synthesis of Reliable Organisms From Unreliable Components”. In: *Automata Studies*. Ed. by C. E. Shannon and J. McCarthy. Princeton: Princeton University Press, 1956, pp. 43–98. ISBN: 9781400882618. DOI: [doi : 10.1515/9781400882618-003](https://doi.org/10.1515/9781400882618-003). URL: <https://doi.org/10.1515/9781400882618-003>.
- [Pip85] Nicholas Pippenger. “On networks of noisy gates”. In: *26th Annual Symposium on Foundations of Computer Science (sfcs 1985)*. 1985, pp. 30–38. DOI: [10.1109/SFCS.1985.41](https://doi.org/10.1109/SFCS.1985.41).
- [Gal91] A. Gal. “Lower bounds for the complexity of reliable Boolean circuits with noisy gates”. In: *[1991] Proceedings 32nd Annual Symposium of Foundations of Computer Science*. 1991, pp. 594–601. DOI: [10.1109/SFCS.1991.185424](https://doi.org/10.1109/SFCS.1991.185424).
- [RS91] R. Reischuk and B. Schmolz. “Reliable computation with noisy circuits and decision trees-a general $n \log n$ lower bound”. In: *[1991] Proceedings 32nd Annual Symposium of Foundations of Computer Science*. 1991, pp. 602–611. DOI: [10.1109/SFCS.1991.185425](https://doi.org/10.1109/SFCS.1991.185425).

- [PS94] Alexander Polishchuk and Daniel A. Spielman. “Nearly-linear size holographic proofs”. In: *Proceedings of the Twenty-Sixth Annual ACM Symposium on Theory of Computing*. STOC ’94. Montreal, Quebec, Canada: Association for Computing Machinery, 1994, pp. 194–203. ISBN: 0897916638. DOI: [10.1145/195058.195132](https://doi.org/10.1145/195058.195132). URL: <https://doi.org/10.1145/195058.195132>.
- [Aha+96] D. Aharonov et al. *Limitations of Noisy Reversible Computation*. 1996. arXiv: [quant-ph/9611028](https://arxiv.org/abs/quant-ph/9611028) [quant-ph]. URL: <https://arxiv.org/abs/quant-ph/9611028>.
- [Gro96] Lov K. Grover. *A fast quantum mechanical algorithm for database search*. 1996. arXiv: [quant-ph/9605043](https://arxiv.org/abs/quant-ph/9605043) [quant-ph].
- [Sho97] Peter W. Shor. “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer”. In: *SIAM Journal on Computing* 26.5 (Oct. 1997), pp. 1484–1509. DOI: [10.1137/s0097539795293172](https://doi.org/10.1137/s0097539795293172). URL: <https://doi.org/10.1137/s0097539795293172>.
- [AB99] Dorit Aharonov and Michael Ben-Or. *Fault-Tolerant Quantum Computation With Constant Error Rate*. 1999. arXiv: [quant-ph/9906129](https://arxiv.org/abs/quant-ph/9906129) [quant-ph].
- [BM06] H. Bombin and M. A. Martin-Delgado. “Topological Quantum Distillation”. In: *Phys. Rev. Lett.* 97 (18 Oct. 2006), p. 180501. DOI: [10.1103/PhysRevLett.97.180501](https://link.aps.org/doi/10.1103/PhysRevLett.97.180501). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.97.180501>.
- [Rom06] Andrei Romashchenko. “Reliable computations based on locally decodable codes”. In: *Proceedings of the 23rd Annual Conference on Theoretical Aspects of Computer Science*. STACS’06. Marseille, France: Springer-Verlag, 2006, pp. 537–548. ISBN: 3540323015. DOI: [10.1007/11672142_44](https://doi.org/10.1007/11672142_44). URL: https://doi.org/10.1007/11672142_44.
- [Bom10] H. Bombin. “Topological subsystem codes”. In: *Phys. Rev. A* 81 (3 Mar. 2010), p. 032301. DOI: [10.1103/PhysRevA.81.032301](https://link.aps.org/doi/10.1103/PhysRevA.81.032301). URL: <https://link.aps.org/doi/10.1103/PhysRevA.81.032301>.
- [Got14] Daniel Gottesman. *Fault-Tolerant Quantum Computation with Constant Overhead*. 2014. arXiv: [1310.2984](https://arxiv.org/abs/1310.2984) [quant-ph].
- [Bom15] Héctor Bombín. “Gauge color codes: optimal transversal gates and gauge fixing in topological stabilizer codes”. In: *New Journal of Physics* 17.8 (Aug. 2015), p. 083002. DOI: [10.1088/1367-2630/17/8/083002](https://doi.org/10.1088/1367-2630/17/8/083002). URL: <https://doi.org/10.1088/1367-2630/17/8/083002>.
- [BMS17] Michael J. Bremner, Ashley Montanaro, and Dan J. Shepherd. “Achieving quantum supremacy with sparse and noisy commuting quantum computations”. In: *Quantum* 1 (Apr. 2017), p. 8. ISSN: 2521-327X. DOI: [10.22331/q-2017-04-25-8](https://doi.org/10.22331/q-2017-04-25-8). URL: <https://doi.org/10.22331/q-2017-04-25-8>.
- [Gro19] Antoine Grospellier. “Constant time decoding of quantum expander codes and application to fault-tolerant quantum computation”. Theses. Sorbonne Université, Nov. 2019. URL: <https://theses.hal.science/tel-03364419>.
- [ABN24] Anurag Anshu, Nikolas P. Breuckmann, and Quynh T. Nguyen. “Circuit-to-Hamiltonian from Tensor Networks and Fault Tolerance”. In: *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*. STOC ’24. ACM, June 2024, pp. 585–595. DOI: [10.1145/3618260.3649690](https://doi.org/10.1145/3618260.3649690). URL: <http://dx.doi.org/10.1145/3618260.3649690>.
- [BMV24] Mitali Bafna, Dor Minzer, and Nikhil Vyas. *Quasi-Linear Size PCPs with Small Soundness from HDX*. 2024. arXiv: [2407.12762](https://arxiv.org/abs/2407.12762) [cs.CC]. URL: <https://arxiv.org/abs/2407.12762>.

- [Ngu24] Quynh T. Nguyen. *Good binary quantum codes with transversal CCZ gate*. 2024. arXiv: [2408.10140](https://arxiv.org/abs/2408.10140) [quant-ph]. URL: <https://arxiv.org/abs/2408.10140>.
- [NP24] Quynh T. Nguyen and Christopher A. Pattison. “Quantum Fault Tolerance with Constant-Space and Logarithmic-Time Overheads”. In: *Proceedings of the 57th Annual ACM Symposium on Theory of Computing* (2024). URL: <https://api.semanticscholar.org/CorpusID:273850180>.
- [Pal+24] Louis Paletta et al. “Robust sparse IQP sampling in constant depth”. In: *Quantum* 8 (May 2024), p. 1337. ISSN: 2521-327X. DOI: [10.22331/q-2024-05-06-1337](https://doi.org/10.22331/q-2024-05-06-1337). URL: <https://doi.org/10.22331/q-2024-05-06-1337>.
- [TKY24] Shiro Tamiya, Masato Koashi, and Hayata Yamasaki. *Polylog-time- and constant-space-overhead fault-tolerant quantum computation with quantum low-density parity-check codes*. 2024. arXiv: [2411.03683](https://arxiv.org/abs/2411.03683) [quant-ph]. URL: <https://arxiv.org/abs/2411.03683>.
- [KZ25] Anirudh Krishna and Gilles Zémor. *Tradeoffs on the volume of fault-tolerant circuits*. 2025. arXiv: [2510.03057](https://arxiv.org/abs/2510.03057) [cs.IT]. URL: <https://arxiv.org/abs/2510.03057>.
- [Li+25] Yiming Li et al. *Poincaré Duality and Multiplicative Structures on Quantum Codes*. 2025. arXiv: [2512.21922](https://arxiv.org/abs/2512.21922) [quant-ph]. URL: <https://arxiv.org/abs/2512.21922>.